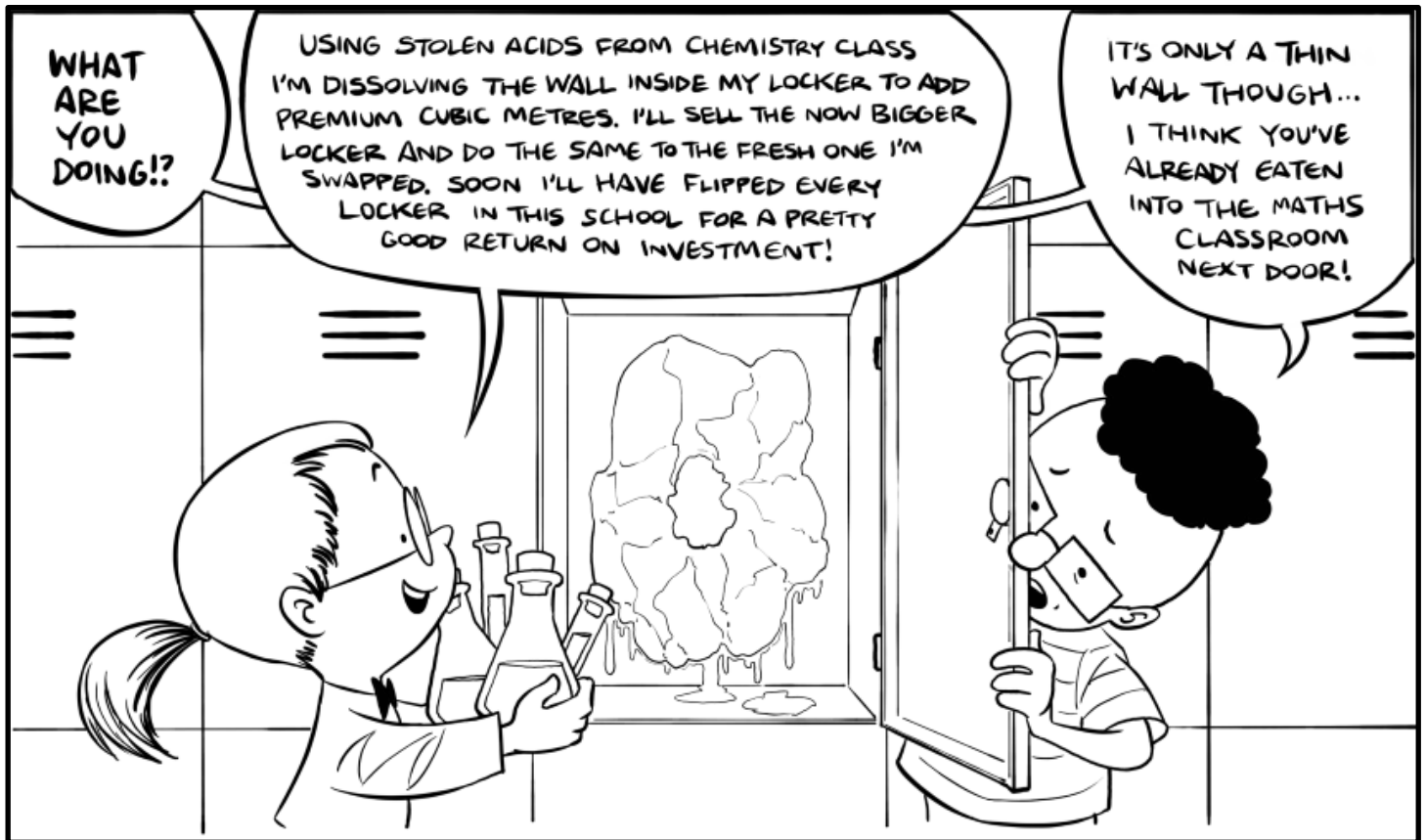


Securities Education Certificate



Exclusive SEC Cartoon by Nicolas Geiseler Toran (Mechanical Engineering, 2022) [7/7]

Lecture 7

Real Estate & Commodities

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Table of Contents

Real Estate	4
Introduction	4
Defining Real Estate	4
Types of Real Estate	4
Residential	4
Commercial	4
Industrial	5
Vacant Land	5
Special Purpose	5
Historical Prices of Real Estate	5
Purchase of Physical Property	5
Homeownership	6
Rental Property	6
House Flipping	6
Calculating Return on Investment	7
Cost Method	7
Cost Method for Homeownership/ House Flipping example	7
Cost Method for Rental Property example	7
Out-of-Pocket	8
Out-of-Pocket for Homeownership/ House Flipping example	8
Out-of-Pocket for Rental Property example	9
Mortgage-Backed Securities	9
Before Mortgage-Backed Securities	10
After Mortgage-Backed Securities	10
Financial Crisis	11
Real Estate Investment Trusts	12
REIT Qualification	12
Types of REITs	12
Equity REIT	12
Mortgage REIT	12
Hybrid REIT	13
Classification of REITs	13
Evaluation of REITs	13
Valuating REITs	13
Quantitative methods	14
Funds from Operation	14
Adjusted Funds from Operation	15
Net Asset Value	15
Qualitative methods	15
Types of property	15
Types of Tenants	Error! Bookmark not defined.
Geographic Location	16
Occupancy and Rent	16

Commodities	17
Introduction	17
What are commodities?	17
Classification	17
Examples of commodities	17
The Commodity Market Players	18
Hedgers	18
Buy-side hedging	18
Sell-side hedging	18
Speculators	18
Why Invest in/Trade Commodities?	18
Strong long-term fundamentals of commodities	19
Population Growth	19
Urbanisation	19
Industrialisation	19
Hedge against inflation	19
Safe haven of value	19
Inelastic nature	20
Commodity diversification	20
Commodity Focus: Crude Oil	20
Where are the crude oil reserves?	21
Where is crude oil produced?	22
Where is crude oil consumed?	22
Who are the crude oil importers and exporters?	23
What are the historic prices of crude oil?	23
How do you invest/trade in commodities?	24
Futures	24
Forwards	24
Options	24
Indexes through ETFs and Mutual Funds	24
Stocks and MLPs	25
CTAs and CPOs	26
Important Commodities Market Indicators	27

Real Estate

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Introduction

The real estate market is worth \$217 trillion – making it the world's most important asset class (by value). In fact, global real estate is a more valuable asset class than all the stocks, shares and bonds combined. Real estate is something we all experience on a day-to-day basis and will continue to play a vital role throughout our lives. Whether you are renting property or own property yourself, it is important to understand the basics behind the real estate industry. This section of the securities education certificate will take a deep dive into multiple different types of real estate investment including topics such as purchase of physical property, mortgage-backed securities, and real estate investment trusts.

Defining Real Estate

We will begin by defining real estate.

The basic definition is as follows:

Real Estate – A property consisting of land or buildings.

However, we can take this a step further to include any changes made to the land, so we can define real estate as:

Real Estate - Land or any permanent improvements to the land whether natural or man made.

Some examples of what can be considered as permanent improvements to the land include water, trees, minerals, buildings or fences. We also need to differentiate between real property and personal property when defining real estate. Real property is property that is permanently attached to the land such as the things listed above. However, personal property is not included in the definition, hence things such as jewellery or vehicles are not considered as real estate.

Types of Real Estate

Having considered the definition of real estate we can break this down further into the different types of real estate.

Residential

Residential real estate is property used for habitation purposes. Examples of residential real estate include Family residences, townhouses, detached houses and apartments.

Commercial

Commercial real estate is any property that is used exclusively for business purposes. Properties such as office buildings, hotels, shopping centres and parking facilities all come under commercial real estate.

Industrial

Any property used for manufacturing, production, distribution, storage, or research and development. Examples of industrial real estate would include factories, power plants, warehouses, and data centres.

Vacant Land

This can simply be defined as any undeveloped land that has no buildings on it and land that is not being used. This includes raw land, natural fields, and forests.

Special Purpose

Finally, special purpose land is any property used by the public. Government buildings, public libraries, cemeteries, and places of worship all come under special purpose real estate.

Historical Prices of Real Estate

Before we start our dive into the various types of real estate investments, I would like to highlight the previous performance of the industry by looking at the average UK house price since 1990. With the major exception of the 2008 financial crisis, house prices tend to demonstrate continuous year on year growth and can be one of the safest assets to invest in. Despite the onset of COVID-19 causing house prices to take a small hit, the vaccine rollout and reduction in pandemic concerns cause prices to accelerate to reach an all-time high in 2022. Factors such as supply of housing, interest rates, and general market sentiment are the main contributors to this price rise.

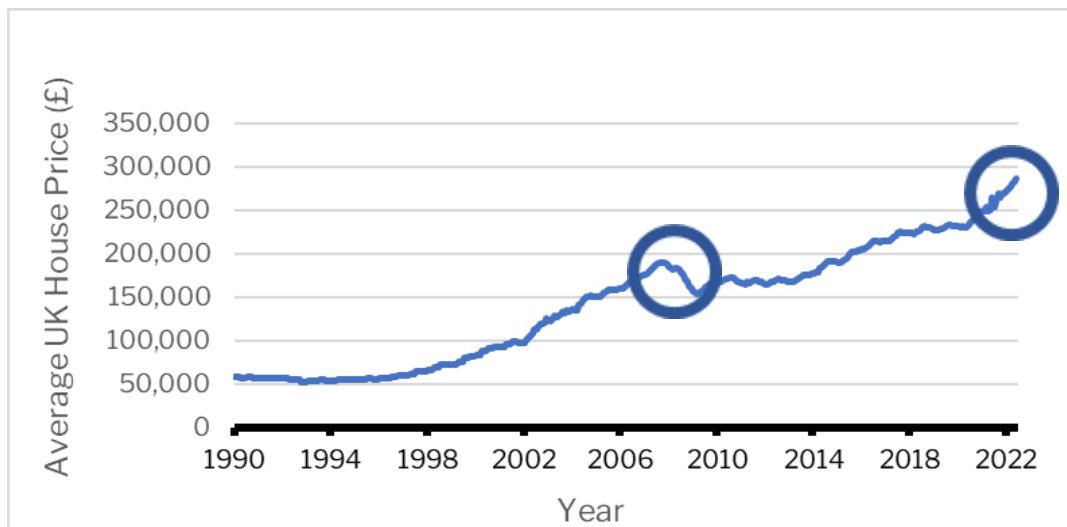


Figure 1: Average UK House Price from 1990 - 2022

Purchase of Physical Property

The first method of real estate investment we are going to cover in this part of the SEC is the purchase of physical properties. This involves the concept of home ownership, rental properties, and house flipping. The way that profit is generated from these methods involve revenue from rent/ leases or appreciation of property value or a combination of both.

Homeownership

Purchase of residential property for the owner to live in.

Owning a home is something we all strive towards and is one of the most common forms of real estate investment. Despite having a place to live being the major reason for purchase of residential real estate, homeowners benefit from the appreciation of the assets value too. In addition to this, diversification of the homeowner's portfolio and the fact property can be bought using leverage (mortgage) are also considered as advantages. Furthermore, the inflation hedging characteristics of real estate can help homeowners reduce the impact an overheating economy has on their personal finances. As we have seen over the past year, high inflation causes the cost of living to rise with the rental market seeing a considerable increase in rental prices. This extra cost is not something homeowners face thus reducing the impact high inflation has on them. However, purchase of residential real estate does not come without its disadvantages. Price increases year on year tend to be quite small with an average percentage increase of 10.1%. There is no guarantee that the property will remain at the price it was bought at since geographical properties can have a considerable impact on property prices. Many first-time buyers are realising the large initial capital required to purchase a property with the average down payment on a mortgage of 12%. This factor also makes property markets illiquid since large amounts of paperwork and fees can be associated with selling/ buying property.

Rental Property

Purchase of property with the interest of leasing it out to tenants.

Rental property (also called buy to let property) involves the owner of the property becoming the landlord and generating income through the collection of rent. This rental income is considered as regular passive income which can diversify the owner's income sources. Like homeownership, the landlord benefits from the appreciation of value and the fact that the property can be bought using leverage (mortgage). Again, rental property can also have negatives associated with it. An obvious disadvantage is the fact a landlord is unable to generate income when the property is vacant. In addition to this, management of tenants can be tedious and any damage caused by tenants will be an extra cost incurred by the landlord.

House Flipping

Purchase of a property with the intention of holding it for a short period of time (3 – 4 months) and quickly selling it.

This method of real estate investment is far less common than the two outlined above due to the increased knowledge required and risk associated with it. House flipping can be done in two ways; repair and update or hold and resell. Repair and update involves the landlord purchasing the property, paying for renovations or any repairs needed, then selling the property off at a higher price. The profit generated by this is equal to the difference in resale and purchase price minus the cost of any repairs and updates done. In essence the investor is hoping that the changes made to the property will add more in value than the cost required to carry out the works. Hold and resell involves an investor purchasing a property then simply holding the property for a certain time period with the hope that property prices increase. Once the price has increased, the investor will sell the property above the initial purchase price. Both these methods come with the benefit of tying up capital for a short period of time and offering quick returns. However, strong market knowledge is required since accurate valuations are vital to ensure this method is profitable. Furthermore, there is always the risk that the property market cools, and the property cannot be sold at a higher price than what it was purchased at.

Calculating Return on Investment

Return on investment (ROI) – the money earned on an investment after all expenses are deducted.

ROI is a key figure used by real estate investors to determine whether a particular investment is worth the costs associated and is a metric used to compare one strategy to another. In this course we are going to cover two methods: the cost method and out-of-pocket method. Each of these methods will then be applied to both homeownership and rental properties. For rental properties, return on investment tends to be calculated on an annual basis.

Cost Method

ROI using the cost method is found by dividing the investment gain on a property by the property's initial cost. It also makes the key assumption that the property is purchased outright with cash and without financing. The equation below is used to calculate ROI using the cost method:

$$\text{Cost Method ROI (\%)} = \frac{\text{Net Investment Gain}}{\text{Initial Cost}} \times 100$$

Cost Method for Homeownership/House Flipping Example

Take a property with initial value £100,000 to which the homeowner adds repairs and improvements costing £50,000. This increases the market value of the house to £200,000 and we are going to calculate the return on investment.

First step is to calculate the net gain from the investment:

$$\begin{aligned} \text{Net Investment Gain} &= £200,000 - (£100,000 + £50,000) \\ &= £150,000 \end{aligned}$$

Next, we need to calculate the initial cost:

$$\begin{aligned} \text{Initial Cost} &= £100,000 + £50,000 \\ &= £150,000 \end{aligned}$$

Finally, ROI can be worked out by substituting net investment gain and initial cost into the ROI equation:

$$\begin{aligned} \text{Cost Method ROI (\%)} &= \frac{£50,000}{£150,000} \times 100 \\ &= 33.3 \% \text{ (to 3 sf)} \end{aligned}$$

Cost Method for Rental Property Example

Take a property with initial value £100,000 to which the homeowner adds repairs and improvements costing £10,000. Monthly expenses (bills/maintenance) costs £250 and the rent charged by the landlord is £1200 per month.

Calculating annual net gain from the investment:

$$\begin{aligned} \text{Annual Net Investment Gain} &= (12 \times £1,200) - (12 \times £250) \\ &= £11,400 \end{aligned}$$

Next, we need to calculate the initial cost (same as for homeownership/ house flipping):

$$\begin{aligned} \text{Initial Cost} &= £100,000 + £10,000 \\ &= £110,000 \end{aligned}$$

Finally, ROI can be worked out by substituting net investment gain and initial cost into the ROI equation:

$$\begin{aligned} \text{Cost Method Annual ROI (\%)} &= \frac{£11,400}{£110,000} \times 100 \\ &= 10.4 \% \text{ (to 3sf)} \end{aligned}$$

Out-of-Pocket

This method for determining ROI uses leverage to purchase property instead of purchasing the property outright. Out-of-pocket ROI is preferred by most investors as it results in a higher ROI. The equation below can be used to calculate ROI using the out-of-pocket method. One thing to highlight is that the equations for Out-of-Pocket method vary slightly for homeownership/ house flipping when compared to rental property. The equation for each is highlighted in each example.

Out-of-Pocket for Homeownership/House Flipping Example

$$\text{Out - of - Pocket ROI (\%)} = \frac{\text{Current Equity Position}}{\text{Final Value of Property}} \times 100$$

Take a property with initial value £100,000 to which the homeowner adds repairs and improvements costing £10,000. However, this time instead of purchasing the property outright, the owner decides to finance the property with a mortgage which requires a down payment of £20,000.

First we will calculate the original out-of-pocket expenses:

$$\begin{aligned} \text{Original Out - of - Pocket Expenses} &= £20,000 + £50,000 \\ &= £70,000 \end{aligned}$$

In order to calculate the current equity position, we subtract the original out of pocket expenses from the final value of the house.

$$\begin{aligned} \text{Current Equity Position} &= £200,000 - £70,000 \\ &= £130,000 \end{aligned}$$

Finally to calculate the ROI we simply divide the current equity position by the final market value:

$$\begin{aligned} \text{Out - of - Pocket ROI (\%)} &= \frac{£130,000}{£200,000} \times 100 \\ &= 65\% \end{aligned}$$

Notice how the return on investment when calculated using this method is much higher than the cost method. Hence why this method is preferred by investors.

Out-of-Pocket for Rental Property example

$$\text{Out - of - Pocket ROI (\%)} = \frac{\text{Annual Rental Income} - \text{Annual Expense}}{\text{Original Out - of - Pocket Expense}} \times 100$$

Take a property with initial value £100,000 to which the homeowner adds repairs and improvements costing £10,000. Monthly expenses (bills/maintenance) costs £250 and the rent charged by the landlord is £1200 per month. This time the property is financed using a mortgage which requires an initial downpayment of £20,000. The annual principal + interest payment is £7,000.

Calculating annual net gain from the investment:

$$\begin{aligned} \text{Annual Net Investment Gain} &= (12 \times £1,200) - (12 \times £250) - £7,000 \\ &= £4,400 \end{aligned}$$

Next, we need to calculate out-of-pocket expense:

$$\begin{aligned} \text{Original Out - of - pocket expense} &= £20,000 + £10,000 \\ &= £30,000 \end{aligned}$$

Finally, ROI can be worked out by substituting net investment gain and initial cost into the ROI equation:

$$\begin{aligned} \text{Annual ROI (\%)} &= \frac{£4,400}{£30,000} \times 100 \\ &= 14.70 \% \text{ (to 3sf)} \end{aligned}$$

Mortgage-Backed Securities

Having discussed one method of real estate investment, the purchase of physical properties, we can move to another method called **mortgage-backed securities (MBS)**.

Before Mortgage-Backed Securities

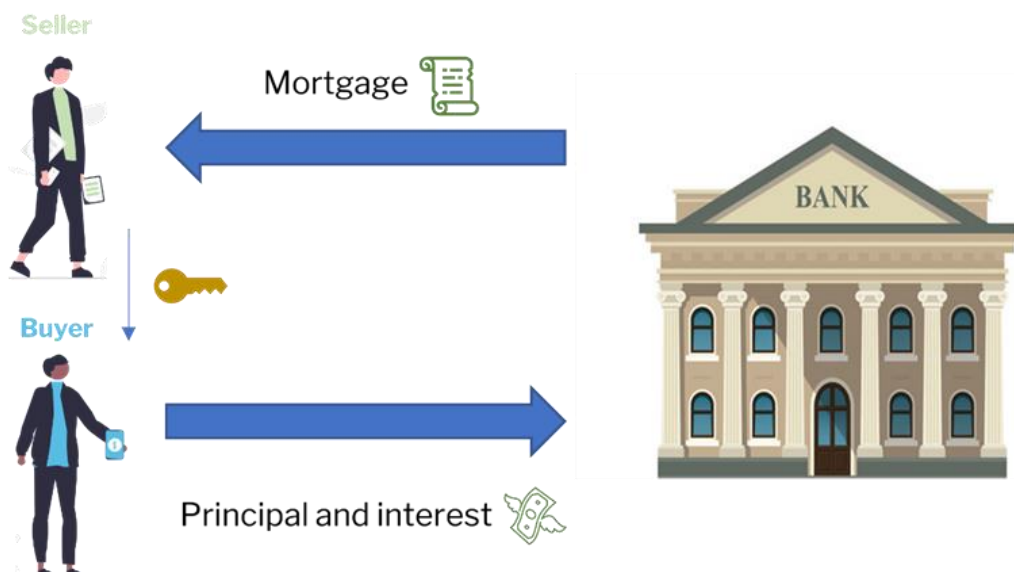


Figure 2: Transactions before MBS

To understand how mortgage-backed securities work we will need to take a step back and investigate the transactions involved between parties when purchasing a property using leverage. If we completely ignore the idea of an MBS, three main parties are involved: the seller, the buyer and the bank. Firstly, the buyer and seller will agree on a price for which the property can be purchased, and the buyer will go to the bank and apply for a mortgage. Once the buyer's credentials have been approved, a legally binding contract will be signed, and the transaction process begins. The seller receives the payment from the bank and hands over the keys to the house. This is where the seller leaves the picture and the transactions occur purely between the buyer and the bank. Each month (or whenever is agreed on the contract) the buyer will make a payment of the principal, amount that you borrow from the lender, and interest, the cost of borrowing money. For the duration of the mortgage (typically 20 – 30 years), the mortgage is kept on the bank's books and the bank continues to receive the principal and interest each month.

The issue with this way of managing mortgages is that it ties up the bank's resources and capital for the duration of the loan. This results in an opportunity cost, the potential benefits lost out when choosing an alternative over another, due to the lack of funds available to the bank to use in other transactions (e.g. other mortgages).

After Mortgage-Backed Securities

To avoid the problem outlined above, the concept of a mortgage-backed (MBS) security was established with the first MBS issued in 1968. A mortgage-backed security is a type of bond representing an investment in a pool of real estate loans.

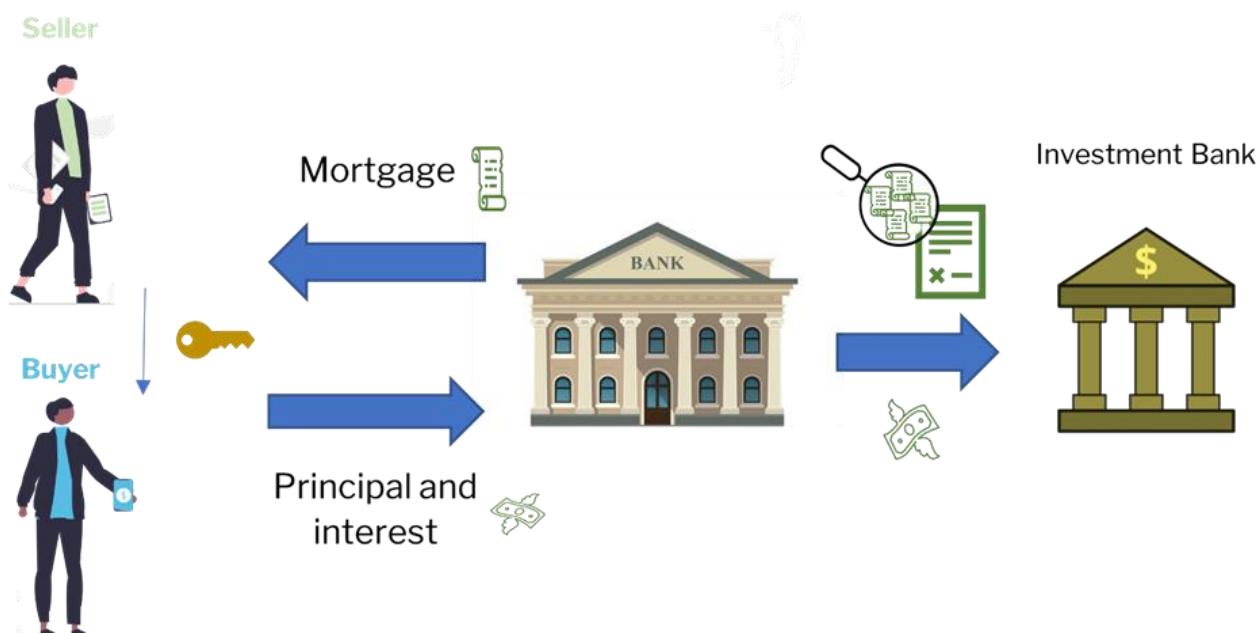


Figure 3: Transactions with MBS

With mortgage-backed securities, the initial part of the transaction remains the same. The seller receives the payment for the house and the buyer continues to pay principal and interest each month. The difference is what the bank decides to do with the mortgage. This time instead of keeping the mortgage on the bank's books it decides to bundle lots of mortgages, usually mortgages with the same investment rating, and sells this to investment banks in the form of a bond. The investor receives the principal and interest paid to the bank by the seller. By doing so, the investor who buys a mortgage-backed security is essentially lending money to home buyers. The MBS turns the bank into an intermediary between the homebuyer and the investment industry. This process frees up the bank's capital (possibly allows the bank to issue more mortgages) and allows the bank to make money from originating and servicing the mortgage as well as any other associated fees. In addition to this, the investors make money from the interest, principal and changes in the value of the MBS. This process works well for all concerned as long as all parties involved do what they are supposed to do. This means that the bank keeps to reasonable standards when granting mortgages, homeowners continue to make payments on time and credit rating agencies that review MBS perform due diligence.

Financial Crisis

Now that we understand the concept of MBS, we can take a look at the central part it played in the 2008 financial crisis. With house prices continuing to increase year on year and the growing demand for mortgages and MBS, banks began lowering their lending standards to cope with demand. By doing so, a large number of people were granted mortgages despite low credit ratings and high risk of default (subprime mortgages). Despite this, the banks knew that with house prices rising rapidly if a person were to default, they could recoup any losses by taking the house as collateral and selling it on the market. As subprime borrowers began to default on loans, the housing market was flooded with homes causing prices to decrease rapidly resulting in the housing collapse seen from 2006 to 2008. As a result of the housing collapse, people began walking away from their mortgages because their homes were worth less than their loans. The sheer amount of non-payments meant that many MBSs, and collateralized debt obligations based on pools of mortgages were very overvalued. Huge losses piled up as institutional investors and banks tried and failed to unload bad MBS investments. Credit in the economy tightened leaving many banks on the brink of insolvency. Lending as a whole was disrupted so badly that the entire economy was on the brink of collapse.

Real Estate Investment Trusts

Now that we have discussed two out of the three major methods of real estate investment, what's left to cover is the concept of **real estate investment trusts or REITs** for short. The real estate sector is known for having very high barriers to entry with the average house price at £299,249 and average commercial real estate cost at £2,190 / m². With such high barriers to entry, it can be difficult to finance real estate investments meaning that most people will miss out on the benefits from investing in real estate.

A real estate investment trust is a company that owns, operates or finances income producing real estate. The concept of a REIT was established by Congress in 1960 to make the benefits of income producing real estate accessible to anyone and everyone. Today, approximately 145 million Americans own REIT stocks either directly or indirectly through mutual funds, ETFs or target date funds. Investing in a well-diversified REIT is like owning a small part of all of the REIT's assets such as office buildings and hotels. It allows easy entrance to the real estate market and can be invested in with very small amounts of money so it is easy to finance.

REIT Qualification

In order to be classified as a REIT, the company must follow certain rules outlined and enforced by the Securities and Exchange Commission (U.S.) or Financial Conduct Authority (U.K.). When established as a REIT, the company does not need to pay corporation tax on any profits arising from the property business within a REIT.

A company must follow the rules outlined below to be classified as a REIT:

- Invest in at least 75% of total assets in real estate, cash, or U.S. Treasuries.
- Derive at least 75% of income from rents, interest on mortgages that finance real estate property, or real estate sales.
- Pay out a minimum of 90% of taxable income in the form of shareholder dividends each year.
- Be an entity that is taxable as a corporation
- Be managed by a board of elected directors or trustees
- Have at least 100 shareholders after its first year of existence
- Have no more than 50% of its shares held by five or fewer individuals

If, and only if, the company follows these key rules can it be classified as a REIT and reap the benefit of being such a status.

Types of REITs

REITs can be split into three types: equity, mortgage and hybrid REITs.

Equity REIT

An Equity REIT is where the REIT itself owns income producing real estate. This is the most common type of REIT where revenue is generated from rent paid by the REIT's tenants. An example of a REIT is Prologis, which mainly focuses on industrial and logistics investment and owns over 1 billion square feet in 19 different countries. An investor can own a small part of each one of these buildings by investing in Prologis REIT. This outlines the benefit REITs bring to small investors especially with exposure to the commercial real estate sector.

Mortgage REIT

Mortgage REITs finance real estate owners as opposed to owning income producing real estate themselves. This type of REIT can finance owners directly, through loans, or indirectly, through investment in mortgage-backed securities. Its main revenue source is from interest charged on loans. Annaly Capital Management is the

world's largest mortgage REIT with over \$89 billion in total assets. Its primary focus is on borrowing via short term security repurchase agreements and reinvesting proceeds into asset backed securities. The dividend yield of the company is currently at 10.68% which means that for every £100 pounds invested, you will receive £10.68 in dividends each year. Such high dividend rates is a key feature of REITs and one that we will understand later.

Hybrid REIT

Finally a Hybrid REIT, as the name suggests, is a combination of both an equity and Mortgage REIT. This means that it owns income producing real estate and also finances real estate owners both directly and indirectly. Link REIT is the largest hybrid REIT in Asia. The company manages a diverse portfolio of retail facilities, car parks, offices, and logistics centres.

Classification of REITs

In addition to the type, REITs can be classified depending on how their shares are traded.

Public REIT – Shares for public REIT are listed on a national securities exchange and are regulated by the Securities and Exchange Commission (US) or Financial Conduct Authority (UK).

Publicly Non-Traded REIT – This does not trade on a national securities exchange but is regulated by the Securities and Exchange Commission (US) or Financial Conduct Authority (UK).

Private REITs – This type of REIT is not traded on a national securities exchange or regulated by either the SEC or FCA. Shares of private REITs tend to be quite illiquid.

Evaluation of REITs

Having understood the concept of REITs and established the different types of REITs we can look into the advantages and disadvantages in investing in real estate investment trusts.

The main advantage of a REIT, and also the reason why they were created, is the low entry price to the real estate market (commercial real estate in particular). It allows everyday people to gain access to an industry that was previously reserved for large institutional investors. In addition to diversifying an investment portfolio, the high liquidity of publicly traded REIT shares means that all kinds of investors can benefit from a real estate investment. Due to the fact that a REIT needs to distribute 90% of its taxable profits as dividends, it can be a source of stable low risk cash flow for the investor.

Despite the numerous advantages, there are also some negatives associated with an investment in a REIT. The key benefit of high dividends also means that there is very little money left over to reinvest into the company and therefore REITs tend to grow at a much lower pace when compared to alternative investments. Due to this characteristic, REIT shares are best purchased with a long term time horizon in mind. Another disadvantage is the fact that dividend yield is taxed as regular income unlike corporation tax. Although you as the investor gain access to the real estate sector, you have very little control over the returns and performance of your investment and you are paying for management fees in addition to the real estate purchase price when investing.

Valuating REITs

With REITs being such an easy way to invest in the real estate market, we need to establish some methods both quantitative and qualitative to understand what makes investing in one REIT better than another.

Quantitative methods

One of the best metrics to use when evaluating most companies is the price to earnings ratio (P/E). The P/E ratio can be used to compare companies within a particular sector and can indicate if a company is overvalued (high P/E ratio) or undervalued (low P/E ratio). However, it is NOT a good metric to use to evaluate a REIT and we will explore why below.

$$P/E \text{ Ratio} = \frac{\text{Share Price}}{\text{Earnings per Share}}$$

If we break down how earnings per share is calculated:

$$\text{Earnings per Share} = \frac{\text{Net Income} - \text{Dividends}}{\text{Outstanding Shares}}$$

Looking further into how Net Income is derived:

$$\text{Net Income} = \text{Revenue} - \text{Expenses}$$

	Year 1	Year 5	Year 10
Balance sheet, Assets:			
Real estate (gross assets)	1,000,000	1,000,000	1,000,000
(-) Minus accumulated depreciation	0	250,000	500,000
Net real estate (net assets)	1,000,000	1,000,000	1,000,000
Income statement			
Revenues	200,000	200,000	200,000
Expenses			
Operating Expenses	100,000	100,000	100,000
Interest Expense	40,000	40,000	40,000
Annual depreciation	50,000	50,000	50,000
Total expenses	190,000	190,000	190,000
Net income (revenue – expenses)	10,000	10,000	10,000
Funds from operation (FFO)	60,000	60,000	60,000
(FFO = net income + depreciation + etc)			

Figure 4: Example Balance Sheet & Income Statement

The issue we have when it comes to REITs is the way expenses are calculated on the income statement. When calculating expenses for the income statement, any assets the company owns must be depreciated and is written as the annual depreciation in the income statement. This is then taken away from revenue to get the company's net income. The issue is properties tend to appreciate, not depreciate. Therefore, calculating price to earnings ratio in the method explained above will give an inaccurate indication of the company's performance since the amount of depreciation included in the expenses will be very high for a company holding predominantly real estate assets.

Funds from Operation

In order to mitigate the problem associated with using price to earnings ratio we use a metric called Funds From Operation (FFO). The equation for calculating FFO is show below:

$$\text{FFO} = (\text{Net Income} + \text{Depreciation} + \text{Amortisation} + \text{Losses on Property Sales}) - (\text{Gain on Sales of Property}) - (\text{Interest Income})$$

Depreciation and amortisation are the expensed portions of a company's tangible (physical) and intangible assets for the year. FFO adds these expenses back into net income to determine the companies actual incoming cash/ revenue from operations. Any losses on sales of business property, normally including long-term assets, are considered as one-time and non recurring so are not a part of normal operations. For the same reason any gains on property sales are taken away from FFO. Finally, we subtract interest income as it is not generally a regular part of the business's normal operation thus should not be included in the funds from operation calculation. Funds from operation can be used to give an idea of cash generated by the REIT and can be converted into metrics such as P/FFO which can be used to compare different REITs with each other.

Adjusted Funds from Operation

Depreciation and amortisation are the expensed portions of a company's tangible (physical) and intangible assets for the year. FFO adds these expenses back into net income to determine the companies actual incoming cash/revenue from operations. Any losses on sales of business property, normally including long-term assets, are considered as one-time and non recurring so are not a part of normal operations. For the same reason any gains on property sales are taken away from FFO. Finally, we subtract interest income as it is not generally a regular part of the business's normal operation thus should not be included in the funds from operation calculation. Funds from operation can be used to give an idea of cash generated by the REIT and can be converted into metrics such as P/FFO which can be used to compare different REITs with each other.

$$AFFO = FFO + \text{Rent Increases} - \text{Capital Expenditure} - \text{Routine Maintenance Amounts}$$

AFFO gives a full picture of a company and can help shareholders get an idea of any residual cash flow left to pay out as dividends. It also takes into consideration additional costs and income sources giving an overview of future capability to pay dividends.

Net Asset Value

NAV or Net Asset Value assesses the fair market value of a REIT's real estate assets. The use of NAV is to bypass book value in favour of providing a more accurate estimation of the actual market value for a REIT's holdings. Calculation of NAV can be very subjective since it requires you to determine the value of each of the company's assets. One way of doing this is capitalization of the operating income, basing it on market rates. A cap rate for the present market is determined and used to divide a property's operating income, with the resulting amount being the estimated market value. The market value minus any mortgage liabilities gives the NAV. This is another metric that can be used to compare similar sized REITs.

Qualitative methods

In addition to the quantitative methods outlined above we can use qualitative methods to ensure the stability of a REIT in a range of economic climates.

Types of Property

Diversity of the property sector a REIT owns is very important to its success. For example, a REIT which is diversified in the residential, retail, hospitality and office buildings may have more stability than a REIT purely focused in hospitality for example. If the economy were to experience a shock, the well diversified REIT is likely to have a more resilient performance over the single market REIT. As well as this, knowing the sectors a REIT is involved in can be a good way to predict future performance in various economic climates. The diagram below shows the breakdown of the properties held by AEWU REIT which is a UK based REIT.

AEWU REIT SECTOR WEIGHTINGS

■ Industrial ■ Offices
■ Retail Warehouse ■ Standard Retail
■ Leisure

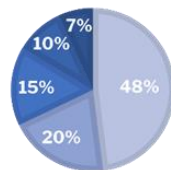


Figure 4: AEWU REIT Sector Weightings

Types of Tenants

By exploring the types of tenants that lease, property owned by the REIT can be very important since the tenant's performance can sometimes have a significant impact on the price of a REIT. The key points to look out for is the company having large reliable tenants but also a large pool of tenants so that the REIT is not adversely affected if one tenant were to have sub-par performance. If we take a look at the diagram below we can see that RioCan, the second largest REIT in Canada has a very diverse range of tenants as well as large reliable tenants such as Walmart and Shell.

Retailer Category	% of Rent	Key Brands (ii)
Grocery / Pharmacy / Liquor	19.9%	COSTCO, Walmart, Loblaws, SHOPPERS, Rexall, LCBO
Essential Personal Services	14.3%	BMO, Canada Trust, Scotiabank, Shell, ROGERS
Specialty Retailers	11.9%	SEPHORA, PETSMART, SPORTCHEK, Michaels, staples
Value Retailers	10.6%	WINNERS, Marshalls, DOLLARAMA, DSW
Furniture & Home	8.9%	STRUCTUBE, HomeSense, BED BATH & BEYOND, The BAICK, LOWE'S
Other Personal Services	8.0%	GoodLife FITNESS, LA FITNESS, Bliss Hair & Beauty
Quick Service Restaurants	7.1%	Tim Hortons, McDonald's, ACW, Five Guys

Figure 5: RioCan REIT Tenant Breakdown

Geographic Location

The geographic location of the company's real estate portfolio can have a huge impact on how a REIT performs. For example, just because one area is facing a property recession, other areas of the country or world may be experiencing a property boom. Therefore, in order for a REIT to perform well through a multitude of economic climates, geographic diversification is a very important factor. The ideal situation would be for a company to be geographically diversified both within a country (e.g. a range of states in America) and internationally (own property in multiple countries).

Occupancy and Rent

Finally, it is important to look into the occupancy and rent characteristics of the REIT. The occupancy rate measures how much of a company's property is vacant as a percentage of total property owned. High occupancy rates (95%+) is a good sign for a company. As always this is very dependent on the sector, for example the ideal occupancy rate for a hotel may be very different to the ideal rate for warehouse space. In addition to this, the average lease length of a REIT's tenants can be a great measure for the stability of the REIT. In general long lease lengths demonstrate tenants have confidence in the company but also provide a secured source of income for the future.

Commodities

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Introduction

What are commodities?

Commodities can formally be defined as **basic goods possessing fungibility**.

Not getting very far with this definition are we...

More practically, basic goods are the raw materials that we use as humans to live our daily lives and keep up with our standards of living. Hence, You can imagine from examining your lifestyle that these goods are numerous. From the natural gas used to heat your shower in the morning, the rice you eat for lunch, to the steel used to construct the very building in which you are – these are all examples of the basic goods that we are talking about. These goods are usually either seen as a safe store of value or are integral to the functioning of our modern industries.

As for fungibility, it designates the interchangeability of the good produced, meaning the good is the same whether it is produced by producer A or producer B. For example, gold is a commodity as it is a basic good/raw material/natural resource that is fungible – a gold bar from Peru is that same as a gold bar from Australia of the same purity. This allows for a singular price to be attributed to the same commodity of different provenances and reassures the investor in what exactly they are buying into when investing in a commodity security (more on that later).

Classification

Commodities can be grouped into hard commodities and soft commodities. Hard commodities are extracted from the earth and include energy and metals, whilst soft commodities are 'grown' and include agricultural commodities as well as livestock and meat.

The metal commodities can be sub-categorised into precious metals and base metals depending on the nature through which they derive their value

Examples of commodities

Energy: Crude oil, Natural gas, Coal

Precious metals: Gold, Silver, Platinum, Palladium

Base metals: Steel, Nickel, Aluminium, Copper

Agricultural: Corn, Rice, Wheat, Coffee, Sugar, Cocoa

Livestock and Meat: Live Cattle, Feeder Cattle, Juicy Bacon (I'm not joking – they're also known to traders as frozen pork bellies. It has the electronic ticker of GPB on the CME Globex)

Okay... enough with my obsession that some traders are getting paid seven figures to hoard bacon on the daily. Now that we know what commodities are, let's look at why we should even be interested in investing in commodities in the first place.

NOTE: You may hear some people refer to currencies, bonds and equities as 'commodities'. In this lecture, I will only be referring to non-financial commodities as... well, *commodities*.

The Commodity Market Players

It should first of all be noted that there are two types of players that invest in the commodities market: hedgers and speculators.

Hedgers

Hedgers are commercial players who are either a producer or consumer of a commodity. Hedgers enter the commodities market in order to reduce price volatility on the commodities that they buy/sell as part of their core business. Hedgers are thus protected from the daily fluctuation of prices by 'locking-in' a price months or even years in advance with a futures contract.

Buy-side hedging

An airline company's business depends on the price of fuel. To protect itself from the erratic fluctuations, it buys fuel contracts in bulk, in advance.

Sell-side hedging

A farmer wants to guarantee the return that he/she will get from selling the corn that he cultivates. He is satisfied with the current market price of corn but is afraid that corn prices may drop in a month when he will have harvested his corn. He thus sells a corn contract at a predetermined price and date to protect himself from the risk.

The buyer of a futures contract is referred to as the *holder*, and the seller as the *underwriter*.

Speculators

Speculators are also referred to as traders/investors. These players don't need the commodities they are trading. They buy and sell contracts with the sole purpose of profiting from the price fluctuations of the commodities market.

Examples include savvy investors like you all (or Ahmad and Oscar). Or, just your typical commodities trader at a desk in Morgan Stanley.

Thus, hedgers try to protect themselves from the price fluctuations while speculators try to profit from it.

Why Invest in/Trade Commodities?

Historically overlooked by investors, the commodities bull-run of the 90s and 00s attracted a lot of attention to its market as speculators sought to profit from it. Numerous investment instruments were thus created which made it easier for investors to buy into the commodities market.

There are numerous reason why investors have been increasingly interested in commodities as an asset class:

Strong long-term fundamentals of commodities

Population Growth

It is a well-known trend that the population on this planet is booming. Yes, we did just reach 8 billion people on the 15th of November 2022! The UN forecasts show that by 2055, we will have reached 10 billion people. These people all need resources like you and me – food to eat, clothes to wear, buildings to live in, fuel for heating and transport – and this all requires raw materials. As we compete to get our hands on the world's finite resources (peacefully or not), the fundamentals of supply and demand mean that the prices of commodities will only increase as demand increases.

Urbanisation

Urbanisation is another significant demographic phenomenon. In the early 20th century, according to the UN, less than 15 percent of the population lived in cities, this has since increased to 55 percent in 2022 and is forecast to reach 68 percent in 2050.

Urbanisation is highly significant for commodities because people living in cities, through their urban lifestyle, consume a lot more than their rural counterparts. Think more miscellaneous shopping, more imported consumption rather than local production (e.g. food, textiles), more fuel needed for transport etc. In addition, the expansion of cities requires new roads, new rails, new buildings which all consume resources.

Industrialisation

Lastly is the large-scale industrialisation that is happening across developing countries. Though many developing countries are industrialising at lightning speed, five of them are known as the BRICS (Brazil, Russia, India, China, and South Africa) which account for a majority of the world's economic growth. As these countries fight to secure supplies of key resources for industrialisation (e.g. steel, oil, gas, aluminium), this will only continue to put an upward pressure on commodity prices.

Hedge against inflation

There are two main reasons why people invest their money: to make money provide returns and to beat inflation.

Inflation eats into your investments and reduces the real purchasing power of your assets. If you have been reading the news these past few months, you will have noticed that we have entered a period of high inflation, with inflation currently sitting at 11.1 percent in the UK, compared to the central bank's target of 2 percent (Nov 2022). Central banks usually tame inflation through increased interest rates which causes economic activity to fall and stocks to underperform as costs increase for businesses. Commodities are one of the only asset classes that benefit from inflation, as ironically, the increase in their prices is usually a large contributor to inflation itself.

Safe haven of value

In times of turmoil and market volatility, some investors turn to precious metals such as gold and silver for financial safety. These investors either view precious metals, especially gold, as a safe haven of value, or anticipate an increase in its price due to past correlation. Recent situations include the covid pandemic, the war in Ukraine, and the loss in value of currencies. More on that in the gold section below.

Inelastic nature

In economics, elasticity is defined as the influence price has on supply and demand. Elasticity seeks to quantify the effect that incremental change in price has on supply and demand. Elastic goods react to price changes whereas inelastic ones don't (as much). Being necessity goods, many commodities fall into the latter category of inelasticity. Even as the price of commodities increase, many people will continue to desire them as they will still need that gas for heating, and that food to eat.

Commodity diversification

Between 1971 and 2007, the commodity markets posted positive returns in six of the years where the returns of the S&P 500 index were negative. Due to the different fundamentals that drove the prices of commodities compared to stocks and bonds, commodities were deemed a hedge against general market downturns and a necessary asset to achieve a diverse portfolio. This theory was largely true until substantial speculative funds flowed into the commodities market in the 2000s, causing as a result a correlation to appear between the equity and commodities markets – with both rising when global prospects improved, and monetary policies were eased. A more refined theory since, explains that both markets are correlated when driven by demand changes (e.g. both dropping with reduced demand during pandemic), but uncorrelated when driven by supply changes (the war in Ukraine's effect on energy and grain commodities as opposed to the S&P 500).

Now that we are convinced as to why commodities make good investments, let's look more closely at some specific commodities that we can invest in. We will be examining the big boyz in each commodity category.

Commodity Focus: Crude Oil

The energy commodities are mainly composed of the three ways through which humans still obtain the majority of their energy: crude oil, natural gas and coal. As part of the energy commodities category we will delve into the king of commodities: crude oil.

Providing 40% of the world's total energy needs whilst being the most traded commodity on the market, crude oil truly is the king of them all. Its production value and central role in our modern economy only cements that title.

Before investing in any commodity, just like any other financial security, one must fully understand the fundamentals that drive its prices. For commodities, here are several questions that you need to ask yourself:

- Where are the reserves located?
- What companies are exploiting it?
- What are its uses?
- What has been the historic performance
 - ◆ What is the outlook for the future considering current supply and demand trends?
- How is it affected by macroeconomics?

In Figure 1, you can see a graph from BP showing the world consumption of energy and how oil is a big driver in the way the world functions as we see today.

So, let's answer some of these questions for our case on crude oil.

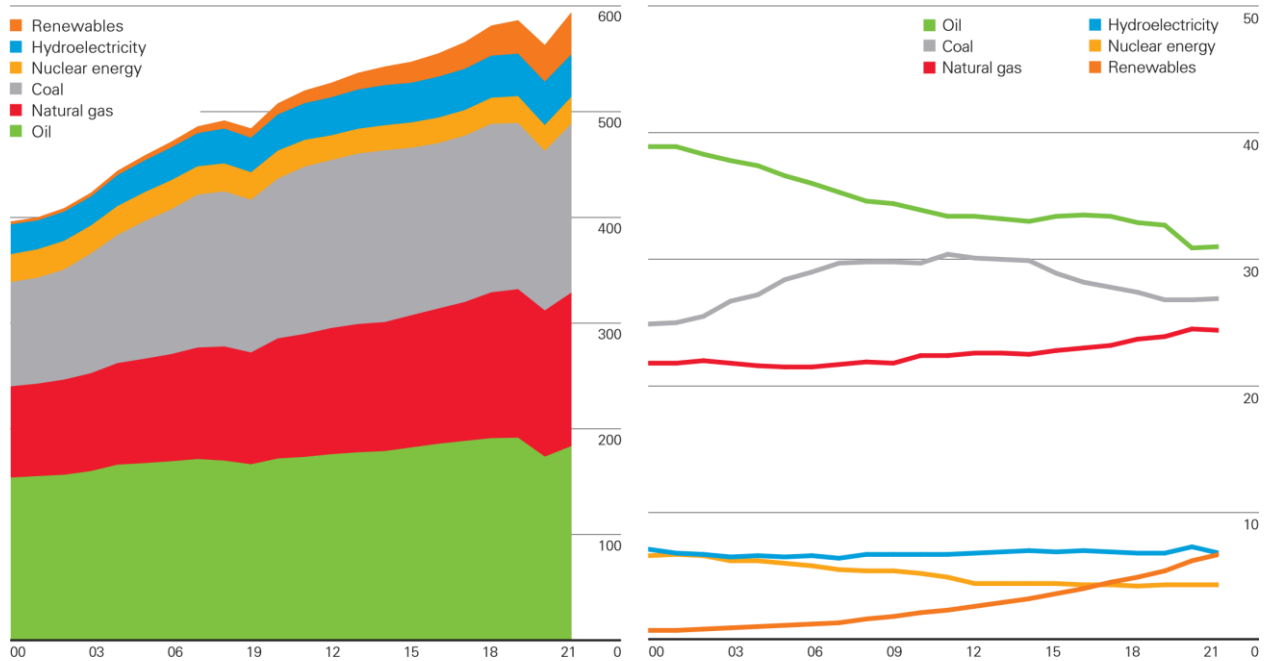


Figure 1: World Consumption of Energy (BP Statistical Review, 2022)

LEFT World consumption measured in Exajoules (10^{18}J)

RIGHT Shares of global primary energy, given as percentages (%)

Where are the crude oil reserves?

Proven Crude Oil reserves (as of 2016)

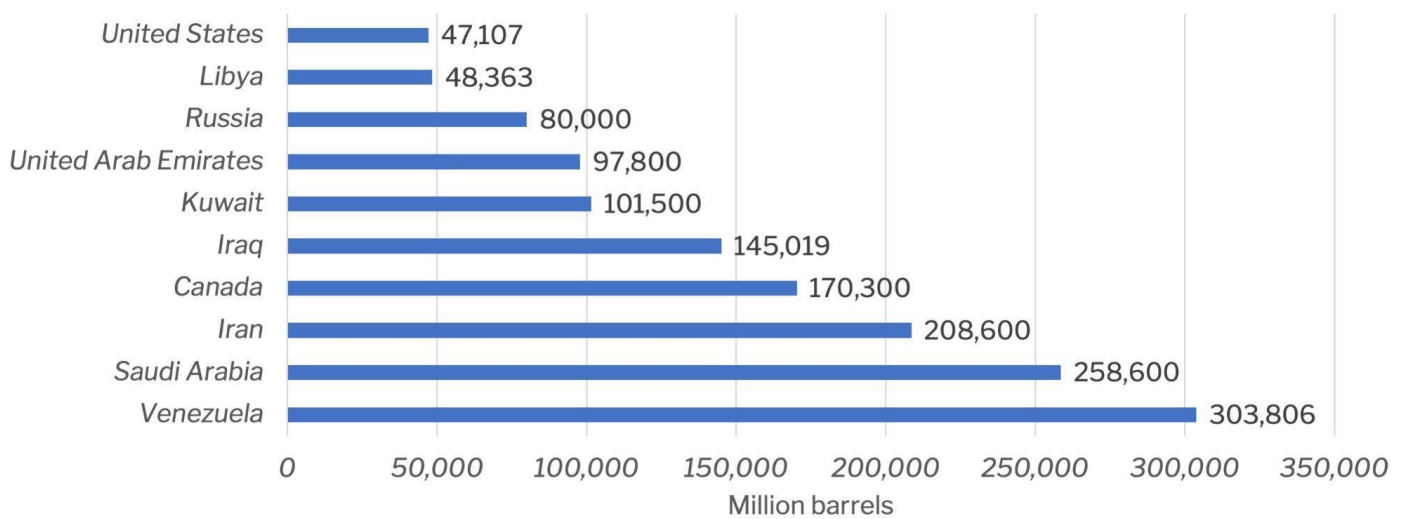
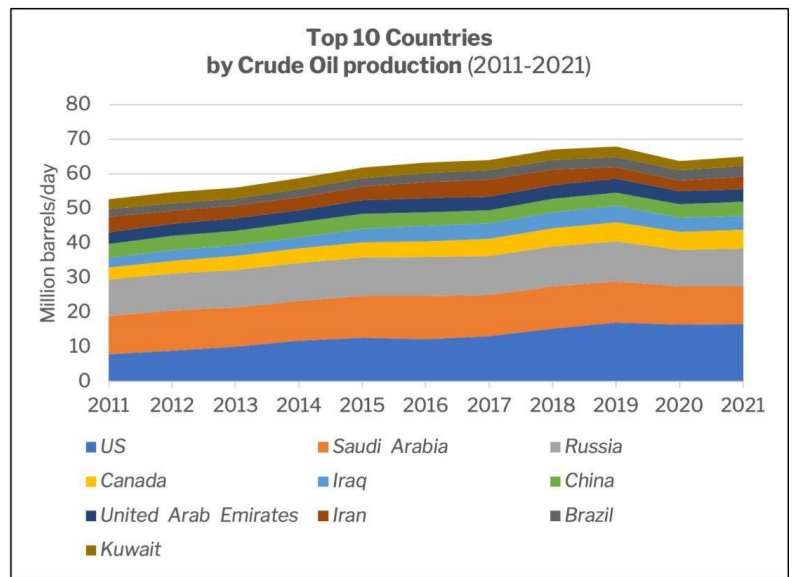
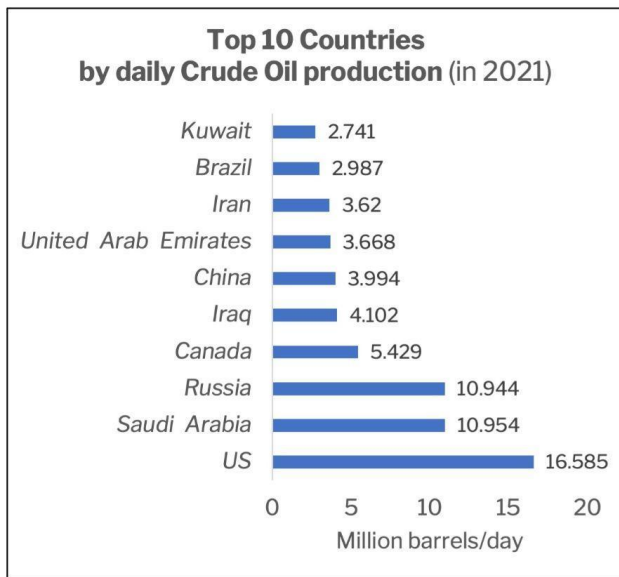
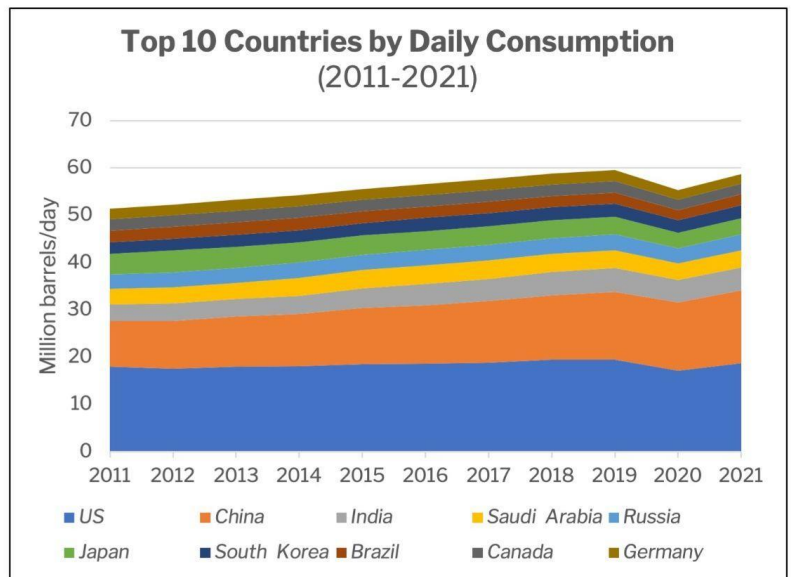
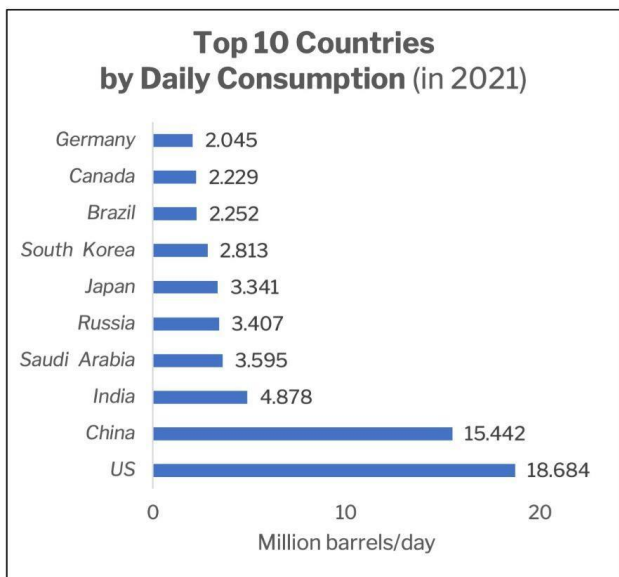


Figure 2: Top 10 Countries by Proven Oil Reserves (US Energy Information Administration, 2016)

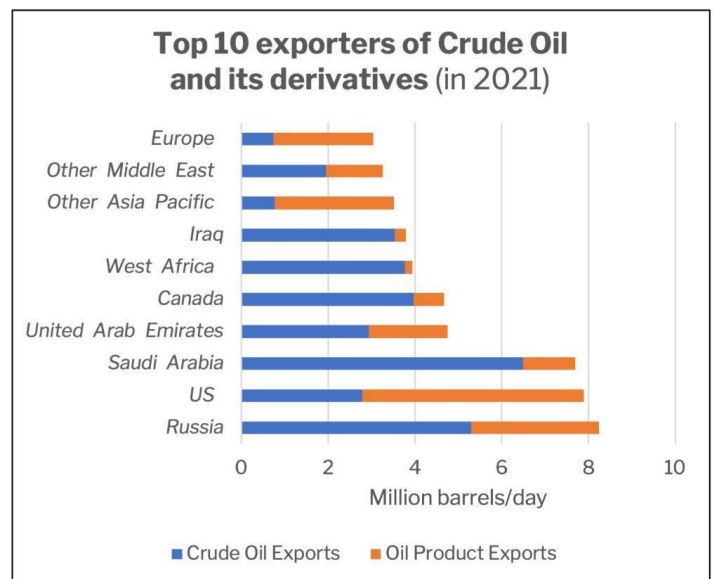
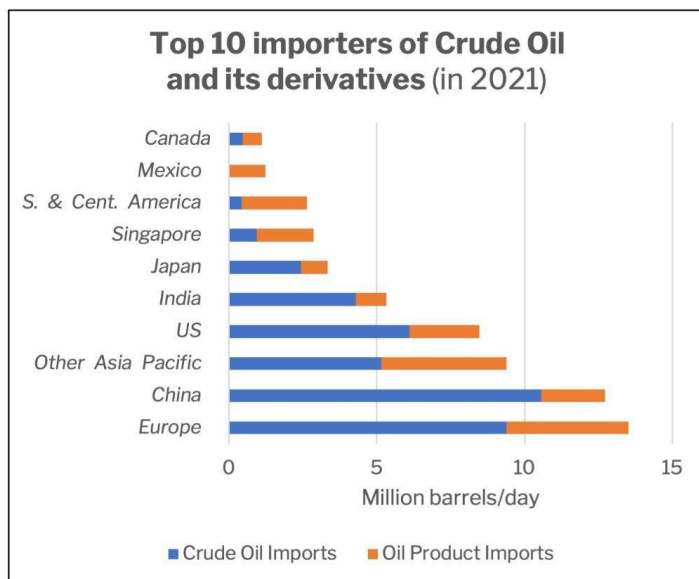
Where is crude oil produced?



Where is crude oil consumed?



Who are the crude oil importers and exporters?



What are the historic prices of crude oil?

Crude oil prices 1861-2021

US dollars per barrel
World events

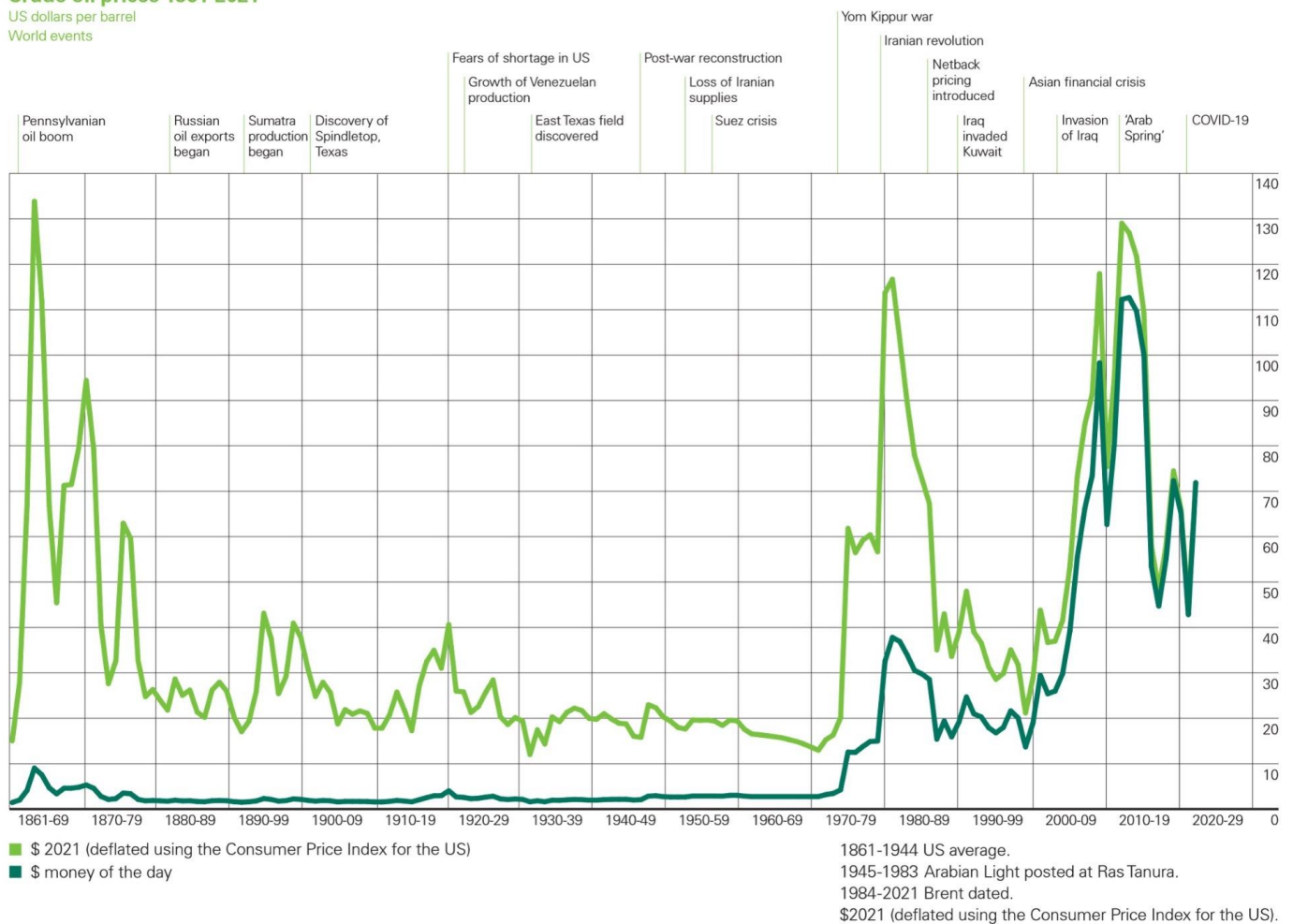


Figure 3: A History of Crude Oil Prices 1861-2021 (BP Statistical Review, 2022)

How do you invest/trade in commodities?

There are many investment vehicles through which investors can gain exposure to commodities. As I am aware that you have already been exposed to most of these, you can skip sections you are already familiar with. However, they are still here in case you would like to read this document independently of the others.

Futures

Futures contracts are the most direct way of getting exposure to commodities. They are used to assume or shift price risk and can be settled either via delivery or offset. Futures allow you to purchase a specific underlying commodity at a mutually agreed upon price at a specific time in the future, hence the name.

In addition, the quality, quantity and delivery location of the commodity are also specified and standardised. Futures contracts trade on designated commodity exchanges such as the Chicago Mercantile Exchange (CME) and the London Metal Exchange (LME). The exchange provides liquidity and transparency to all market participants through its regulatory framework and thus increases investor confidence.

Forwards

The nature of the futures market means that only 20% of market activity happens through the exchange. Forwards are just like futures in the sense that they represent a contractual obligation to purchase/sell a specific underlying commodity at a mutually agreed upon price at a specific time in the future. However, the transaction does not happen through the exchange but rather through the Over-the-Counter (OTC) market – directly between two market participants. This contract is understandably much more used by hedgers as being outside the exchange and free from its standardisation of contracts, it provides the flexibility of tailoring the quality, quantity and delivery location of the commodity to the specific needs of the producer/consumer.

Yeah mate, don't get involved here, unless your grandma has suddenly planted too many potatoes in her backyard and wants to get a good cash bonus on top of her retirement when the harvest comes in next year.

Options

Options are just like futures, being standardised and trading through exchanges. However, one big difference distinguishes the two. While the holder of a futures contract has both the right and the obligation to fulfil the contract's obligations, the holder of an option has the right but **NOT** the obligation to fulfil the option's obligations. For this privilege, the holder of an option pays a premium.

The technically correct way of thinking about it would be that commodity options give you the option to purchase/sell commodity futures. Just like stock options give you the option to purchase/sell stock.

We won't go into specific examples here as it has already been covered in previous parts of the course.

Indexes through ETFs and Mutual Funds

Commodity indexes track a basket of underlying commodities futures contracts, with the composition and weighting of each differing from one to another. The price of an index is essentially governed by an algorithm that seeks to mimic the returns of the underlying contracts. Indexes are a good way of determining how the commodity market is doing in general due to the broad exposure it has. Thus, as well as being a model of a low-risk investment vehicle for the investor that is unsure of which specific commodity to invest in, they are also a benchmark and indicator of the commodity market as a whole.

NOTE: Note: Due to futures contracts having an expiration date, these are rolled regularly within an index. Contracts such as the crude oil futures are rolled monthly because they expire every month. However, other contracts that possess expiration dates on only certain months of the year are rolled on according to the available monthly contract trade.

Here are some commodity Indexes to look into:

- The S&P Goldman Sachs Commodity Index (S&P GSCI) is one of the most closely watched indexes in the market. Tracking 24 commodity futures contracts, the index is production-weighted – assigning different weights to different commodities proportional to their current global production quantity, e.g. crude oil has a higher weighting than cocoa.
- The Reuters/Jefferies Commodity Research Bureau Index (CRB) is another commodity index that is closely followed by institutional investors. Tracking a basket of 19 commodities, this index is unique in that it uses a tiered methodology to distribute weightings. The components of the weightings are reviewed annually.
- The Dow Jones – AIG Commodity Index places a premium on liquidity and chooses commodities based on their production value. The index is one of few that places a weighting floor and ceiling such that no commodity may account for less than 2 percent of more than 33 percent of weighting. This greatly reduces the weighting of crude oil to half of what it usually is in other commodity indexes and allows for greater exposure to other commodities.

It should be noted that you cannot invest into a commodity index directly. Just like you can't directly invest in other indexes such as the famous S&P 500 as these are trackers with no value behind them. You can however invest in a fund that tracks the performance of an index. This can be either through an exchange-traded fund or a mutual fund.

A quick reminder (although you will be able to find them again in the notes for Lecture 3):

- ETFs – a new breed of investment that tracks the performance of a fund with the convenience of trading a stock.
- Mutual funds – an actively managed fund that purchases securities with pooled investor capital.

Alternatively, you could just own the futures tracked by index or even more simply, by ironically adding a layer of complexity to our financial instrument (structurers love it), investors can also invest in **index futures**. You guessed it! An index future is a futures contract that tracks the performance of an index (that tracks the performance of specific commodity futures...). These are, however, more capital intensive as futures tend to trade for quite large sums of money due to the quantity of a commodity they represent.

Stocks and MLPs

By buying shares of a company involved in the commodities industry, you can gain indirect exposure to its market. Continuing with our main example of crude oil, the commodity needs to be found, drilled, refined, and distributed throughout its lifecycle. Thus any company involved in these activities would provide an investment exposure to the price of crude oil.

If you are interested in investing in companies involved in the production, transformation, and distribution of commodities, one of the best ways to do so is to invest in a master limited partnership (MLP). The advantage being the tax structure that allows for higher cash payouts.

MLPs are public entities that trade on public exchanges. Just like a public company, an MLP issues units (shares) that can be freely purchased by the public, which is why they are also termed publicly traded partnerships (PTP). This public partnership is run by a general partner for the benefit of the unit holders who are also called limited partners.

MLPs benefit from the tax advantage of only being taxed once. This pass-through tax structure means that all earnings from the partnership are transferred untaxed to investors who then pay income tax on the distribution received. This is in contrast to corporation dividends that are taxed both at the corporation and investor level, allowing for MLPs to give higher payouts.

Avoiding tax, this all sounds very nice, but be careful, however! Investing in companies also exposes you to additional corporate risk: managerial, administrative, operational...

CTAs and CPOs

Now, if you're really a hands-off investment guy and would rather trust someone else (which may after all know a bit more than you) with your hard-earned money, you can go to a commodity trading advisor (CTA) or a commodity pool operator (CPO).

CTAs are just basically conventional stockbrokers who specialise in the futures market. They are qualified to help you open a futures account, help you develop a commodities investment strategy, as well as trading futures contracts on your behalf. Each CTA will have their own investment strategy and philosophy and it is important to have done due diligence before trusting anyone with your capital.

Another way to get access to commodity futures is to join a commodity pool. This designates a pool of funds that trade commodities futures. This pool is managed by a commodity pool operator (CPO), who just like the CTA needs to be licensed to do so.

Investing in commodity pools has three main advantages:

1. **Diversification.** Futures contracts cost a lot, mainly because of the quantity of the commodity they represent, as you saw above in the crude oil section. With more funds available, more futures can be purchased to thus create a more diverse commodity portfolio.
2. **Leverage.** Once again as discussed above, one of the main characteristics that make commodity futures so unique is the leverage that it allows for. More funds, means much more leverage can be taken on, and thus potentially increase returns by manyfold.
3. Reassuring considering the massive leverage that is usually taken on, your maximum loss at any given time is only your principal – the entirety of the money you gave the CPO. Now this may sound like a horrible deal but let me explain why it may be an advantage.

When trading with leverage on an individual account, you can purchase securities on margin, i.e. borrow funds (usually from the broker) to fund the purchase of the futures contract. Now what happens when the position entered does the opposite of what you speculated? You get margin called (see Lecture 4 notes). Not only have you lost your principal, but you still owe money on top of that as you also need to pay the broker back for the money he lent you.

CPOs on the other hand are registered as limited partnerships. This means that even if leverage is used to get into bad trades that result in the pool going bust, you will not be personally liable to the debt. Hence, you only lose the amount you initially invested into the pool i.e. your principal.

Now, hopefully, you would have performed the necessary due diligence before trusting a CTA or CPO with your money in order to minimise such situations. Important topics to inquire about should include:

- Track record
- Past disciplinary actions
- Fees (management, performance and miscellaneous)
- Minimum investment sum

Important Commodities Market Indicators

Simply looking at the price charts of commodity futures will not help you in making a decision on whether a commodity is a good buy. In order to assist you in investment decisions, here are several crucial market indicators you should consider:

- Consumer Price Index (CPI)
- EIA Inventory Reports
- Federal Funds Rate
- Gross Domestic Product (GDP)
- London Gold Fix
- Nonfarm Payrolls
- Purchasing Managers Index (PMI)
- Reuters/Jefferies CRB Index
- US Dollar
- WTI Crude Oil